Competency 2: Explores the evolution of computing devices, so as to be able to describe and compare the performance of modern computers.

Competency Level 2.1: Elicits the significant changes occurred in the computers from generation to generation with more emphasis on the evolution of processors.

Number of Periods: 04

Learning Outcomes:

- Categorizes the early calculating aids with examples.
- Describes the generations of computers with their features in a table.
- Categorizes computers in terms of technology, purpose and size with examples.

Contents:

- History of computing
- Early calculating aids
    - mechanical
    - electromechanical
- Electronic age of computing
- Generation of computers
    - 1G, 2G, 3G, 4G and future
- Different types of classifications
    - Technology
        - analog, digital
    - Purpose
        - Special/general
    - Size
        - super, mainframe, mini, micro
        - (Portable devices –smart phones, tablet devices and phablets)

Concepts and terms to be highlighted:

- Evolution of computing devices.
- Generations of computers and their characteristics.
- Classification of computers based on technology, purpose and size.

Guidance for lesson plans:

- Discuss how the calculation was done in early days like BC.
- Show pictures of early calculating aids.
- Discuss computers that are used nowadays and explain how it was in the past.
- Explain the changes made in computers from period to period.
- Discuss appearance, technology, speed, capacity and efficiency of computers that are produced in every time period.
- Discuss different type of computers used in different fields.
- Discuss mobile devices currently available and their usage.

Guidance for assessments and evaluations:

- Ask students to write how characteristics have changed through the generations of computers.
- Divide the class into groups and assign each group one of the computer classification topic and ask them to list examples for them

Quality inputs:

- Internet facility, Presentations, images of computing devices

Reading Materials:

Calculating aids – Pre mechanical era – before 1450

Abacus is considered as the first calculating device in the world used by Chinese around 5000 BC.

Mechanical era – 1450-1840

The Pascaline was introduced by Blaise Pascal in 1642. It could add and subtract two numbers.

The Stepped Reckoner invented by German scientist Gottfried Wilhelm Leibniz completed in 1694. This device is carried out the operations add, subtract, multiply and divide too.

The difference engine, the first mechanical computer was developed by Charles Babbage in 1880.

He presented the concept of Input, Process and Output that is used in modern computers, for the first time. Therefore, he is considered as the “Father of the computer”.

Electro mechanical era – 1840 – 1940

Electronic valve (vacuum tube) invented by Forest in 1906.

The automatic sequence controller (Mark 1) was the first automatic computer invented by Professor Howard Aiken in 1939.

Electronic Era

1st Generation Computers (1940-1956)

The computers of first generation used vacuum tubes as the basic component.

- ENIAC (Electronic Numerical Integrator and Calculator) was the first electronic digital computer designed by John Mauchly and J. Presper Eckert in 1946.
- EDSAC **(**Electronic Delay Storage Automatic Calculator) was the first full size stored program computer developed by Maurice Wilkes in 1947. EDSAC was built according to the von Neumann machine principles.
- EDVAC (Electronic Discrete Variable Automatic Computer) is considered as the first digital computer that could store program was built in 1948.

- UNIVAC (Universal Automatic Computer) was an electrical computer containing thousands of vacuum tubes that utilized punch cards and switches to input data and punch cards to output and store data.

2nd Generation Computers (1956 – 1963)

In this generation, transistors were used as the fundamental building block.

IBM 1620, IBM 7094, CDC 1604, CDC 3600, UNIVAC 1108 were some computers developed in this generation.

3rd Generation Computers (1964 – 1975)

The computers of third generation used Integrated Circuits (ICs) in place of transistors. A single IC contains many transistors, resistors, and capacitors along with the associated circuitry.

IBM-360 series, PDP (Personal Data Processor), TDC-316 were the computers of 3rd generation.

4th Generation (1975 – 1989)

In this generation of computers VLSI (Very Large Scale Integrated) circuits were used. VLSI circuits having more transistors and other circuit elements with their associated circuits on a single chip made it possible to have microcomputers.

Eg: DEC 10, STAR 1000, PDP 11

5th Generation Computers (1989 – present)

The ULSI (Ultra Large Scale Integration) technology is used in this generation resulting in the production of microprocessor chips having millions transistors and other electronic components.

Eg: Desktop, Laptop, Notebook, Ultrabook

Classification of Computers

1. Based on the Technology
   - (i) Analog Computer - An analog computer is a form of computer that handles *continuous* values such as electrical, mechanical, or hydraulic quantities.
   - (ii) Digital Computer - A computer that performs calculations and logical operations with quantities represented as digits, usually in the binary number system.
2. Based on the Purpose
   - (i) Special purpose computer - Computers are designed to handle a specific problem or to perform a specific task.

(ii) General purpose computer – These are designed to perform a range of tasks.

3. Based on Size

(i) Super Computer - The fastest and most powerful type of computers Supercomputers are very expensive and are employed for specialized applications that require immense amounts of computing power. (Eg: **TIANHE-1)**

(ii) Mainframe Computer - A very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously.

(iii) Mini Computer – Mid size computers mainly used as small or mid-range servers operating business and scientific applications.

(iv) Micro Computer –

Desktop – A personal computer sufficient to fit on a desk

Laptop – A portable computer

Palmtop – A hand size computer

Notebook - A portable computer small and thinner than Laptop.

Smart phone – This is a mobile phone used for mobile communication with an operating system and other advanced facilities.

Tablet - A tablet is a wireless, portable personal computer with a touch screen interface. The tablet is typically smaller than a notebook computer, but larger than a smartphone.

Phablet - A *phablet* is a small pocket sized mobile device that is a bit larger than the size of an average smartphone and smaller than tablet. (Eg: Apple 6Plus, Galaxy Note, etc)

Competency 2: Explores the evolution of computing devices, so as to be able to describe and compare the performance of modern computers.

Competency Level 2.2: Explores the functionality of a computer in relation to the hardware and their interfaces.

Number of Periods: 06

Learning Outcomes:

- Identifies hardware peripherals and their relevant interfaces.
- Identifies the advantages of direct entry input devices over keyboard entry input devices.
- Describes the evolution of CPU and its compatibility with motherboard.
- Categorizes the storage devices.
- Briefly explains the features of each storage device.
- Identifies the need for parallel and grid computing.

Contents:

Major hardware components

- Input devices: keyboard entry and direct entry (keyboard, pointing devices, touchpad, remote control, touch screen, magnetic strip reader, barcode reader, smartcard reader, scanner, digital camera, microphone, sensors, graphic tablets, MICR, OMR and OCR readers, video camera, digitizer, webcam)
- Advantages of direct entry input devices over keyboard entry input devices
- Output devices and their features (CRT monitor, TFT monitor, LED monitor, dot matrix printer, inkjet printer, laser printer, 3Dprinter, graph plotter, speakers)
- CPU and its compatibility with motherboard
- Storage devices (fixed internal hard disk, portable external hard disk, magnetic tape, Optical discs(CD ROM/DVD ROM, CD-R/DVD-R, CD- RW/DVD-RW,DVD-RAM, Blu-Ray) flash memory card, mini disk)
- Parallel and grid computing.

Concepts and terms to be highlighted:

- Input devices of computer
- Output devices of computer
- Central processing unit and motherboard
- Storage devices of computer
- Parallel and grid computing

Guidance for lesson plans:

- Define the computer as a system.
- Discuss different hardware components that are used in a computer system.
- Show and demonstrate the components of a computer
- Discus the input devices and direct input methods used in a computer system.
- List output devices of the computer system.
- Explain what are the devices used in computers to store data.
- Discuss about parallel and grid computing.

Guidance for assessments and evaluations:

- Assignment - Divide the class into groups and assign each group one of the following and ask them to prepare a document with examples and, pictures on the usage of it.
    - Input devices
    - Output devices
    - Storage devices
    - CPU and motherboard
- Provide some MCQ related to this topic.

Quality inputs:

- Sample Input devices, output devices, storage devices, CPUs, motherboards, presentations

Reading Materials:

Input Devices

These are used to feed data and instructions to a computer system.

Key board

Pointing devices

Mouse	Touch pad	Remote control	Touch screen

Direct entry input devices

Magnetic stripe reader	Bar code reader	Smart card reader

Image and video input devices

Video Camera	Digital Camera	Webcam	CCTV

Scanning devices

Flatbed Scanner	MICR	OMR	OCR

Graphic tablet	Digitizer	Microphone

Advantages of direct entry input device over key board entry input device

1. Automatically capture data, images or videos. No need human involvement to input data into the system. Therefore, data entry cost is reduced.
2. Accurate – The data can be input without any human errors.
3. Less time – Data input process is more efficient.

Output devices

Monitor

1. CRT Monitor (Cathode Ray Tube Monitor)

   In cathode ray tube monitor, images are produced when an electron beam strikes a phosphorescent surface

2. LCD Monitor (using Thin Film Transistor-TFT)

A display screen made with TFT technology is a Liquid Crystal Display (LCD) that has a transistor for each pixel, LCD monitors are widely used in notebook and laptop computers.

3. LED Monitor (Light Emitting Diode Monitor)
   Flat screen monitor used as a computer monitor or television.

Advantages of LED Monitor

- Less expensive
- More reliable
- Generate low heat and consume less power
- Higher dynamic contrast ratio
- Longer lifespan and less environmental impact

Printer
Printer is used to produce a hard copy output.

1. Dot matrix printer
   It uses a print head that moves back-and-forth, or in an up-and-down motion, on the page and prints by striking an ink ribbon against the paper, much like the print mechanism on a typewriter.
2. Inkjet printer
   In the inkjet printing mechanism, the print head has several tiny nozzles, also called jets. As the paper moves past the print head, the nozzles spray ink onto it, forming the characters and images.
3. Laser printer
   When a document is sent to the printer, a laser beam draws the document on a selenium-coated drum using electrical charges. After the drum is charged, it is rolled in toner, a dry powder type of ink. The toner adheres to the charged image on the drum. The toner is then transferred onto a piece of paper and fused to the paper with heat and pressure.
4. Graphic plotter
   A plotter is a printer that interprets commands from a computer to make line drawings on paper with one or more automated pens.

CPU and its compatibility with motherboard

1. Socket support.
   The CPU has to be compatible with your motherboard's socket.

2. Chipset support.
   Need to know about the chipset to find the CPUs compatible with that chipset. Chipset manufacturers include AMD, Intel and Nvidia.

3. Motherboard wattage support must support the Thermal Design Power (TDP) of a given CPU.

Need to find out the wattage of the motherboard to determine the support from that to the CPU

4. BIOS support.

If necessary BIOS need to be upgraded to support the CPU

Storage Devices

1. Fixed internal magnetic hard disk

Hard disk is a data storage device that uses magnetic storage to store and retrieve digital information using one or more rigid rapidly rotating disks (platters) coated with some magnetic material.

2. External hard disk

An external hard drive is a portable storage device that can be attached to a computer through a USB Port

3. Magnetic tape

Magnetic tape is one of the oldest technologies for electronic data storage on a magnetic surface. Tape has largely been displaced as a primary and backup storage medium, but it remains well-suited for archiving because of its high capacity, low cost and long durability. It is a sequential recording system that is not good for random access. With tape archiving, there is no online copy for quick retrieval, as everything is vaulted for the long term.

4. Optical discs

An optical disc is an electronic data storage medium that can be written to and read using a low-powered laser beam.

Type	Capacity
CD-ROM	650-900 MB
CD-R	
CD-RW	
DVD-ROM	4.7-9.4 GB
DVD-R	
DVD-RW	
DVD-RAM	
Blu-Ray	25-128 GB

DVD-RAM - DVD-RAM is like ordinary Random Access Memory (RAM), it can be repeatedly read, written to, and erased. DVD-RAM discs can be rewritten 100 times more than a DVD-RW.

5. Flash drive

6. Memory card

Flash drives and memory cards use Electrically Erasable Programmable Read Only Memory (EEPROM) technology to store data on one or more semiconductor chips.

Parallel computing

Parallel computing is a type of computation in which many programs or processes are done simultaneously. Large problems can often be divided into smaller ones, which can then be solved at the same time.

In simple terms, parallel computing is breaking up a task into smaller pieces and executing those pieces at the same time, each on their own processor or on a set of computers that have been networked together.

The following example displays the difference between single processor and parallel processor

- Y = (4 x 5) + (1 x 6) + (5 x 3)

On a single processor, the steps needed to calculate a value for Y might look like:

- Step 1: Y = 20 + (1 x 6) + (5 x 3)
- Step 2: Y = 20 + 6 + (5 x 3)
- Step 3: Y = 20 + 6 + 15
- Step 4: Y = 41

In a parallel computing scenario, with three processors or computers, the steps look something like:

- Step 1: Y = 20 + 6 + 15
- Step 2: Y = 41

The main performance characteristic is the increase in speed. If you use a single computer, and it takes X amount of time to perform a task, then using two similar

computers should cut the time taken to perform that same task in half. If you use three, then it should take a third of the time for the same task, and so on.

However, in practical terms, this is not always true. Because:

- Some task might not be divisible.
- Some task may not be able divide equally.
- It is necessary to take overhead associated with splitting the task up also into account.

So, while there is usually a performance increase, that increase does not follow any set formula.

Grid computing

Grid computing is a distributed architecture of large numbers of computers connected to solve a complex problem. In the grid computing model, servers or personal computers run independent tasks and are loosely linked by the Internet or low-speed networks. Computers may connect directly or via scheduling systems. In Grid computing inter-connected computer systems utilize the same resources collectively. Grid computing usually consists of one main computer that distributes information and tasks to a group of networked computers to accomplish a common goal. Grid computing is often used to complete complicated or tedious mathematical or scientific calculations.

Examples of Grid Applications

- Application partitioning that involves breaking the problem into discrete pieces
- Discovery and scheduling of tasks and workflow
- Data communications distributing the problem data where and when it is required
- Provisioning and distributing application codes to specific system nodes
- Results management assisting in the decision processes of the environment
- Autonomic features such as self-configuration, self-optimization, self-recovery, and self-management

The following is an example for a grid application and its usage patterns.

Schedulers

Schedulers are types of applications responsible for the management of jobs, such as allocating resources needed for any specific job, partitioning of jobs to schedule parallel execution of tasks, data management, event correlation, and service-level management capabilities. These schedulers then form a hierarchical structure, with meta-schedulers that form the root and other lower level schedulers, while providing specific scheduling capabilities that form the leaves. These schedulers may be constructed with a local scheduler implementation approach for specific job execution, or another meta-scheduler or a cluster scheduler for parallel executions.

Competency 2: Explores the evolution of computing devices, so as to be able to describe and compare the performance of modern computers.

Competency Level 2.3: Explores the Von- Neumann Architecture**.**

Number of Periods: 06

Learning Outcomes:

- Describes the stored program concept.
- Names the major components of Von- Neumann architecture.
- Describes fetch- execute cycle.
- Briefly describes ALU, CU, Memory (Registers), data and control bus.
- Draws the Von- Neumann Architecture model and name sits components.
- Describes the need of multi-core processors.

Contents:

Von-Neumann Architecture

- Stored program control concept
- Components [input device, output device, memory, processor (control unit and ALU)]

Fetch-execute cycle

Central processing unit (CPU)

- Arithmetic and logic unit (ALU)
- Control unit (CU)
- Memory (Registers)
- Data and control bus
- Multi-core processors

Concepts and terms to be highlighted:

- Basic block diagram of a computer based on the Von Neumann Architecture.
- Fetch execute cycle.
- Multi core processors.

Guidance for lesson plans:

- Discuss about IPO model and stored program concept.
- Introduction of Von Neumann Architecture with block diagram and its components.
- Explain how the instruction is executed by a processor.
- Introduce Fetch execute cycle.
- Define parts of a CPU and functionality
- Discuss the need of multi core processor.

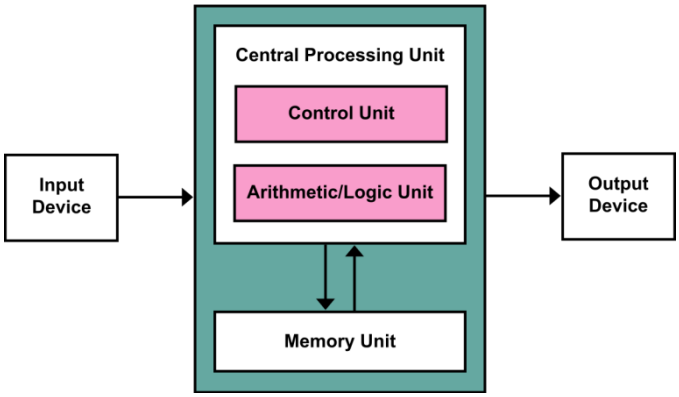
Guidance for assessments and evaluations:

- Ask students to draw and name a block diagram of Von Neumann Architecture.
- Give an assignment to classify the components of a computer

Reading Materials:

Von-Neumann Architecture

Von Neumann Architecture consists of a CPU, memory and input output devices. The program is stored in the memory. The CPU fetches an instruction from the memory at a time and executes it.



(Source; https://en.wikipedia.org/wiki/Von_Neumann_architecture)

Major components of this architecture:

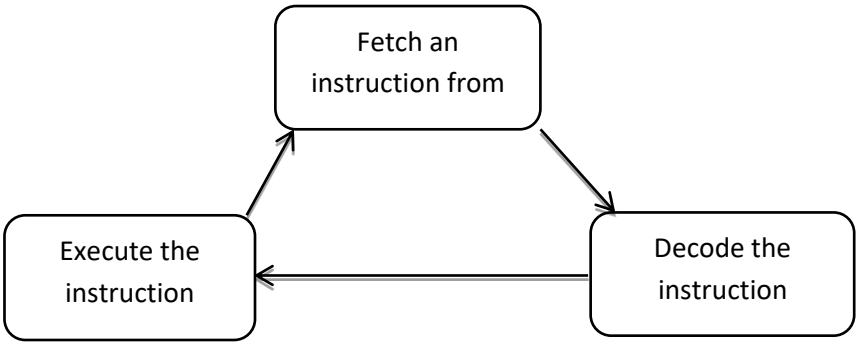
1. Central processing unit
   (i) Control unit (CU)
   This unit controls signals of all devices of a computer system.
   (ii) Arithmetic and logic unit (ALU)
   It carries out mathematical and logical operations.
   (iii) Memory register
   A CPU register is one of a small set of data holding places which is part of the computer processor. A register may hold an instruction, a storage address, or any kind of data

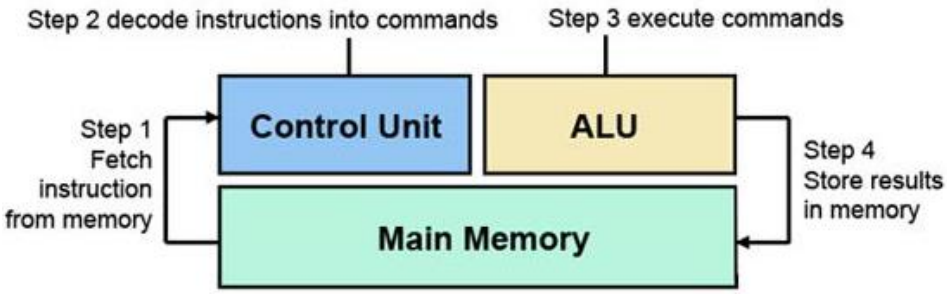
2. Memory
   (i) Primary memory
   (ii) Secondary memory
3. Input device
4. Output device

Data bus
A data bus is a system within a computer or device, consisting of a connector or set of wires, that provides transportation for data.

Control bus
Control bus is used to transmit a variety of control signals to components and devices.

Fetch execute cycle





Multi-core processors

A multi-core processor is a single computing component with two or more independent actual processing units (cores), which are units that read and execute program instructions. Therefore, the single processor can run multiple instructions on separate cores at the same time.

Need of multi-core processor

1. Can be run a program by dividing some parts. So it gets executed fast.
2. It enables parallel programming.
3. To get the high performance from a single machine.

Competency 2: Explores the evolution of computing devices, so as to be able to describe and compare the performance of modern computers.

Competency Level 2.4: Examines PC memory system to identify different types of memory and their main characteristics**.**

Number of Periods: 06

Learning Outcomes:

- Briefly explains the memory hierarchy with a suitable diagram.
- Describes the need for different types of memory and their characteristics.
- Briefly explains the volatile and non- volatile memory.
- Lists volatile and non-volatile memories in computer.
- Describes the characteristics of memory in terms of performance, location, capacity, access method, cost, physical type and physical arrangement of data (bits into words)
- Lists and briefly explains the types and characteristics of ROMs
- Compares and contrasts each type of memory in terms of access time, cost / MB, capacity (typical amount used)

Contents:

- Memory hierarchy
  - Need of memory hierarchy
  - Comparison criteria
    - Physical size/ density of data
    - Access method
    - Access time(elapsed time/delay)
    - Capacity
    - Cost
- Volatile memory and their characteristics
  - Registers
  - Types of cache memory
  - Main memory–RAM
  - Types of RAM
    - SRAM, DRAM, SDRAM
- Non-volatile memory and their characteristics
  - Types of ROMs
    - PROM, EPROM and EEPROM
  - Secondary storage
    - magnetic, optical and flash memory

Concepts and terms to be highlighted:

- Compare memories of computer system with physical size, cost per unit, capacity etc.
- Present the memory hierarchy
- Volatile and non-volatile memory

Guidance for lesson plans:

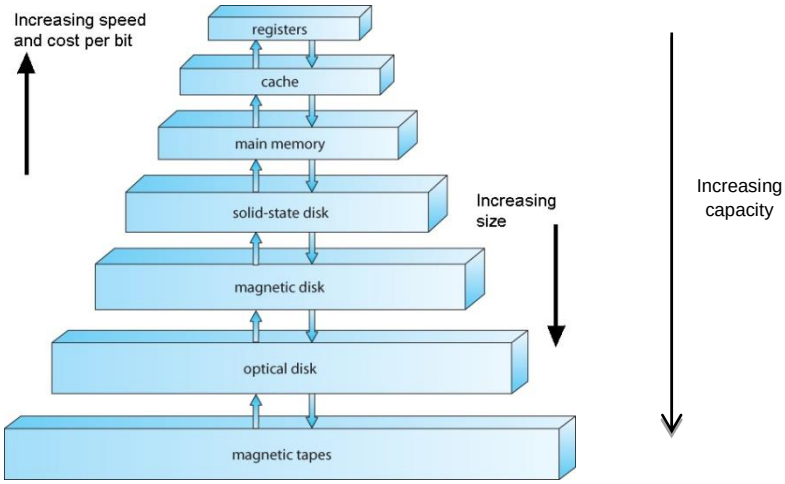
- Draw the memory hierarchy to compare memories.
- List volatile memories and why they needed in computer system.
- Discuss types of cache memory and RAM.
- List non-volatile memories and characteristics of them.
- Discuss types of ROM.
- Explain categorization of secondary storage based on the read/write technology.

Guidance for assessments and evaluations:

- Assignment - Divide the class into groups and assign each group one of the following topic and ask them to list brief description for them.
    - ✓ Register
    - ✓ Cache memory
    - ✓ Random Access Memory
    - ✓ Read Only Memory
    - ✓ Secondary storage

Reading Materials:

Memory hierarchy



Volatile memory

Volatile memory is a computer storage that only holds the data while the device is powered.

Eg: Register, Cache memory, RAM

Cache memory

The cache memory is used to store program instructions that are frequently accessed by software during operation.

Types of cache memories

- Level 1 **(L1)** cache is extremely fast but relatively small, and is usually embedded in the processor chip (CPU).
- Level 2 (L2) cache is often more capacity than L1. It may be located on the CPU or on a separate.
- Level 3 (L3) cache is typically specialized memory that works to improve the performance of L1 and L2. It can be significantly slower than L1 or L2, but is usually double the speed of RAM.

RAM (Random Access Memory)
RAM is the main memory of the computer that holds data for running applications and required data for a computer.
Types of RAM

(i) SRAM – Static RAM
SRAM is random access memory that retains data bits in its memory as long as power is being supplied. SRAM is used for cache memory and register memory.
(ii) DRAM – Dynamic RAM
This type of RAM is continuously refreshed or it will lose its contents.
(iii) SDRAM - Synchronous DRAM
It is a type of memory that synchronizes itself with the computer's system clock.

Non-volatile memory

This is a type of computer memory that has the capability to hold saved data even if the power is turned off.

Eg: ROM, Hard disk etc.

ROM – Read Only Memory

ROM retains its contents even when the computer is turned off. ROM stores essential programs such as the program that boots the computer.

Types of ROM

PROM (Programmable ROM)
It is a memory chip on which data can be written only once. Once a program has been written onto a PROM, it remains there forever.

EPROM (Erasable PROM)
EPROM is a special type of memory that retains its contents until it is exposed to ultraviolet light. The ultraviolet light clears its contents, making it possible to reprogram the memory.

EEPROM (Electrically Erasable PROM)
IT can be erased by exposing it to an electrical charge.

Secondary storage

1. Magnetic storage device
   Magnetic storage is the manipulation of magnetic fields on a medium in order to record audio, video or other data. In main computer storage mechanisms have generally involved a spinning disc or platter and read write heads on an armature. Many types of magnetic storage involve a tape medium ether on a real or in a cassette that is moved by read and write heads.
   Eg: Hard disk, Floppy disk, Magnetic tape

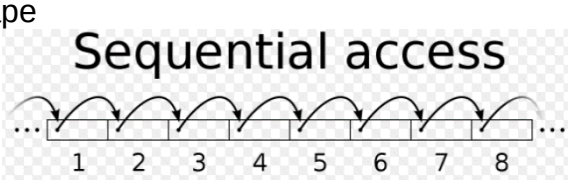
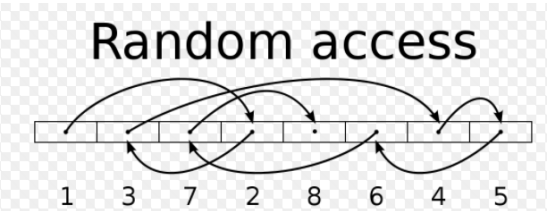
2. Optical storage device
   Optical storage is any storage method in which data is written and read with a laser for archival or backup purposes. Typically, data is written to optical media, such as CDs and DVDs. For several years, proponents have spoken of optical storage as a near-future replacement for both hard drives in personal computers and tape backup in mass storage. Optical media is more durable than tape and less vulnerable to environmental conditions. On the other hand, it tends to be slower than typical hard drive speeds, and to offer lower storage capacities.

   Eg: CD, DVD, Blu-Ray disc

3. Solid state storage
   Solid-state storage (SSS) is a type of computer storage media made from silicon microchips. SSS stores data electronically instead of magnetically, as spinning hard disk drives (HDDs) or magnetic oxide tape do. Solid-state storage can be found in three form factors: solid-state drives (SSD), solid-state cards (SSC) and solid-state modules (SSM). An important advantage of solid-state storage is that it contains no mechanical parts, allowing data transfer to and from storage media to take place at a much higher speed and providing a more predictable lifespan for the storage media. Because there are no moving parts, SSDs produce far less heat than HDDs.
   Eg: Flash drive, Memory card

Memory Access method

(i) Sequential access
Start at the beginning and read through in order.
Eg: Tape



(ii) Random access
Individual addresses identify directly and access the data immediately,
Eg: RAM